

Aggregation, Zero Density, and Branch-Pole Return in Nonabelian Artin–Ihara Models

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Abstract

Nonabelian graph covers offer exact primitive orbits, representation-resolved zeta functions, and finite transfer operators, but these features do not by themselves produce a Hilbert–Pólya model. We isolate three obstructions. First, the canonical product of all nontrivial Artin–Ihara sectors, with regular-representation multiplicities, coarsens each primitive holonomy from its conjugacy class to its element order. An explicit pair of primitive length-ten words in $H(\mathbb{F}_7)$ has identical cyclic directed-bigram counts and is not related by rotation or time reversal, yet has distinct central holonomies; the aggregate merges the pair while a Schrödinger sector separates it. Second, every fixed finite-state, finite-memory determinant with a constant finite-dimensional twist and finitely many positive roof values is an exponential polynomial. A Jensen argument bounds its zero count in a fixed vertical strip by $O(T)$, incompatible with the $\Theta(T \log T)$ Riemann–von Mangoldt law. Third, growing dimension does not rescue the natural Heisenberg congruence tower: primitive exact-conductor Schrödinger blocks have top adjacency eigenvalue tending to four, so nontrivial Bass poles return toward $1/3$ and 1 . At modulus 243 , a rational Rayleigh bound already exceeds $7/2 > 2\sqrt{3}$. These results form a scoped exclusion theorem, not a new Artin factorization or a general no-go result for infinite-rank dynamics.

1 Introduction

A useful dynamical Hilbert–Pólya candidate must do more than put its poles on a circle. Its primitive-orbit data must retain arithmetic information, its determinant must have the right global divisor, and a natural operator must survive the passage from finite computations to one fixed infinite object. Finite graph covers appear to satisfy much of this checklist: nonbacktracking cycles are exact, voltage holonomy records word order, and Artin–Ihara theory resolves the zeta function into finite-dimensional representation sectors [10, 11].

The most obvious unwanted component is the trivial representation. It carries the base graph and its branching poles. One might therefore divide the cover zeta by the base zeta, retain only the nontrivial sectors, and then pass to a congruence tower. This paper tests that construction before attempting any spectral fit. The test reveals a three-way obstruction. Canonical aggregation loses conjugacy-level holonomy; finite representation resolution restores that information but has the wrong zero density; and the natural amenable Heisenberg tower develops exact-conductor nontrivial spectrum arbitrarily close to the trivial spectral edge.

The first obstruction is local and exact. For a primitive base cycle with holonomy $g \in G$, put $o = \text{ord}(g)$ and let x denote its scalar orbit weight. The regular cover divided by the trivial Artin factor contributes

$$(1 - x)(1 - x^o)^{-|G|/o}.$$

This standard consequence of the regular representation depends on g only through o . To show that the loss affects genuine chronological data, we give two cyclically reduced primitive words in $H(\mathbb{F}_7)$ with the same complete cyclic directed-bigram counts. They are neither cyclic nor time-reversal copies, but their ordered products are the distinct central elements z^3 and z^2 . The canonical aggregate identifies them because both elements have order seven. A fixed Schrödinger central character separates the two phases.

The second obstruction is global. A fixed finite-state, finite-memory graph model with finitely many positive roofs has a determinant

$$D(s) = \det \left(I - \sum_{j=1}^r e^{-s\tau_j} B_j \right).$$

The determinant is a finite exponential sum. Its zeros in any fixed vertical strip have count $O(T)$, by a one-page Jensen argument. The nontrivial Riemann divisor has count $\Theta(T \log T)$. Irrationally related finite roofs remove the simple vertical periodicity of the unit-roof graph zeta, but they do not change this density mismatch.

The fixed-dimension theorem deliberately leaves growing towers open. We rule out the uniform Ramanujan/new-sector-gap rescue for the registered natural Heisenberg congruence tower. The four-regular Cayley graph of $H(\mathbb{Z}/3^m\mathbb{Z})$ contains a primitive 3^m -dimensional Schrödinger block

$$H_m = U_m + U_m^* + V_m + V_m^*.$$

A finite-window Rayleigh vector proves $\lambda_{\max}(H_m) \rightarrow 4$. The central frequency is one, so this block does not descend from level $m - 1$. It remains after every one-dimensional sector is deleted. Since the Ramanujan threshold is $2\sqrt{3}$, exact-conductor nontrivial Bass poles eventually leave the graph critical circle and return toward the branching locations.

The paper makes four scoped contributions.

1. We derive the regular-minus-trivial order formula and state precisely which nonabelian information it discards.
2. We exhibit and independently certify a non-dihedral same-bigram chronology witness in $H(\mathbb{F}_7)$.
3. We prove a finite-memory zero-density exclusion that covers both commensurable and finitely incommensurable roofs.
4. We prove branch-pole return in a primitive nonabelian sector of the Heisenberg congruence tower, so deleting all abelian sectors does not rescue this frozen gap mechanism.

The standard graph-covering, finite Heisenberg/Harper, and exponential-sum inputs are explicitly credited. In particular, the Schrödinger block and its near-zero-flux edge behavior are prior objects [2, 3]; the contribution here is the conductor-resolved Rayleigh certificate, its frozen Hilbert–Pólya new-sector-gap application, and the chronology/finite-memory obstruction synthesis. The result does not cover nuclear infinite-rank transfer operators, infinitely many roof values, nonlinear spectral changes, or nonamenable property- τ towers. Those exclusions identify the structural changes that a subsequent construction must make.

2 Background and prior boundaries

Finite graph zeta functions. Hashimoto’s oriented-edge operator and Bass’s determinant formula turn the Ihara Euler product of a finite graph into a finite determinant [1, 5]. For a graph without degree-one vertices, primitive objects are reduced closed oriented cycles modulo cyclic rotation; immediate reversal and tails are excluded. This exact orbit ledger is valuable in a dynamical search because orientation, cyclic multiplicity, and repetition are fixed before computation. It is also restrictive: for a unit roof, the inverse zeta is a polynomial in one variable and its pullback under $u = e^{-s}$ is vertically periodic.

Graph covers and Artin sectors. Regular graph covers admit Artin–Ihara factorizations parallel to finite Galois theory [10, 11]. If a voltage assignment $\alpha : E^\pm(X) \rightarrow G$ satisfies $\alpha(\bar{e}) = \alpha(e)^{-1}$, then a primitive base cycle P carries a conjugacy class g_P . A representation ρ contributes the local factor

$$\det(I - \rho(g_P)u^{|P|})^{-1}.$$

The full regular cover uses every irreducible representation with exponent $\dim \rho$. Sato’s weighted covering formula makes the corresponding prime-splitting law explicit [9]. Our order-only identity in Section 3 is a direct specialization of this standard theory. It is not a new Artin factorization.

Heisenberg graph precedent. Finite Heisenberg Cayley graphs are a natural test because chronological commutators become central area coordinates and the high-dimensional representations produce Harper or Hofstadter matrices. These graph spectra, their irreducible finite-ring Schrödinger blocks, and their Harper matrices were studied directly by DeDeo et al. [3]; the corresponding covers and Artin factors appear in DeDeo et al. [4]. The discrete Heisenberg/Harper spectral edge also has direct precedent [2]. More recent work treats noncommutative pro- p voltage towers and Iwasawa-theoretic special values [7]. We therefore use Heisenberg groups as exact controls, not as newly introduced zeta objects. The new question here is what the canonical subtraction and conductor restrictions can observe.

Exponential sums and zero density. Zeros of exponential sums have a classical theory [6, 8]. We require only an elementary upper bound: a nonzero entire function of finite exponential type obtained from a finite determinant has $O(T)$ zeros in a disk of radius T . We include a self-contained Jensen proof to expose every hypothesis. The comparison target is von Mangoldt’s zero-count law [12],

$$N_\xi(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T).$$

The mismatch is unconditional and concerns the whole divisor, not the location of a short list of zeros.

Scope of the obstruction. A general Hölder roof on a finite-symbol shift need not yield a finite matrix. Nuclear Ruelle operators, infinite weighted graphs, and graph towers can produce genuinely infinite determinants. Our finite-memory theorem does not address those objects. Conversely, showing that every finite truncation has linear density does not control a nonuniform diagonal limit. The Heisenberg theorem separately rules out the uniform Ramanujan/new-sector-gap version of one frozen amenable tower; it does not rule out every renormalized determinant or subtraction scheme and is not used to overextend the finite determinant argument.

3 Canonical aggregation and chronology

Let $Y \rightarrow X$ be a finite regular voltage cover with deck group G . For a primitive base cycle P , let g_P be its holonomy and let x_P be any multiplicative scalar cycle weight. The unweighted case is $x_P = u^{|P|}$, while a suspension roof gives $x_P = e^{-s\tau(P)}$.

Theorem 3.1 (Regular-minus-trivial local factor). *Put $o_P = \text{ord}(g_P)$. The local factor of the regular cover is*

$$\prod_{\rho \in \hat{G}} \det(I - \rho(g_P)x_P)^{-\dim \rho} = (1 - x_P^{o_P})^{-|G|/o_P}.$$

After removing the trivial representation, the canonical local aggregate is

$$\prod_{\rho \neq \mathbf{1}} \det(I - \rho(g_P)x_P)^{-\dim \rho} = (1 - x_P)(1 - x_P^{o_P})^{-|G|/o_P}.$$

Hence this aggregate depends on the holonomy only through its order.

Proof. The regular representation decomposes as

$$\text{Reg}_G \simeq \bigoplus_{\rho \in \hat{G}} (\dim \rho) \rho.$$

Left multiplication by g_P is a permutation of G with $|G|/o_P$ cycles, each of length o_P . Therefore

$$\det(I - x_P \text{Reg}_G(g_P)) = (1 - x_P^{o_P})^{|G|/o_P}.$$

Taking inverse determinants and dividing by the trivial factor $(1 - x_P)^{-1}$ proves the two formulas. $\square$

Remark 3.2 (What collapses). Theorem 3.1 does not say that the scalar zeta loses all chronology. Ordered multiplication can change o_P , and the full product can retain the distribution of orders across primitive cycles. The theorem says that two same-weight base cycles with same-order, nonconjugate holonomies have identical canonical aggregate local factors.

We now make that loss explicit beyond symbol incidence or first-order transition statistics. Write $H_7 = H(\mathbb{F}_7)$ as triples with law

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab').$$

Let $x = (1, 0, 0)$, $y = (0, 1, 0)$, and denote inverses by upper-case letters.

Proposition 3.3 (Same-bigram chronology witness). *The cyclic words*

$$P = \text{XXXyxxxyxYY}, \quad Q = \text{XXXyxxyxxYY}$$

are reduced, primitive, and not related by cyclic rotation or time reversal. They have identical cyclic directed-bigram counts, but

$$g_P = (0, 0, 3) = z^3, \quad g_Q = (0, 0, 2) = z^2.$$

The two holonomies are distinct central conjugacy classes, neither equal nor inverse, and both have order seven.

Proof. Direct inspection excludes adjacent inverse letters, including across the cyclic boundary. Neither length-ten word is a proper power. The nonzero bigram counts of each are

$$xx : 1, xy : 1, xY : 1, XX : 2, Xy : 1, yx : 2, YX : 1, YY : 1.$$

Enumerating the ten cyclic rotations of P and of P^{-1} does not produce Q . Chronological multiplication with the displayed group law gives the two stated central elements. Since $2 \not\equiv \pm 3 \pmod{7}$, the elements are distinct and noninverse. Every nonidentity central element has order seven. $\square$

The canonical aggregate assigns both words the local factor

$$(1 - x)(1 - x^7)^{-49}.$$

Here $x = u^{10}$ for the common unit roof; the same conclusion holds for a weight determined only by the matched one-letter or directed-bigram ledger. We do not assert equality for an arbitrary longer-memory roof. It therefore cannot recover the central coordinate even though the chronological dynamics did. By contrast, in the seven-dimensional Schrödinger sector with $\rho_k(z) = e^{-2\pi i k/7} I$, the local inverse determinants are

$$(1 - e^{-6\pi i k/7} x)^{-7} \quad \text{and} \quad (1 - e^{-4\pi i k/7} x)^{-7}.$$

A resolved sector repairs the observability loss. The next section shows why no fixed finite set of such repairs supplies the required global divisor.

4 A finite-memory zero-density obstruction

Finite incommensurable roofs are often proposed as a way to break the vertical lattice of a finite graph zeta. They do break exact periodicity, but they do not change the growth order of the determinant.

Theorem 4.1 (Finite-memory zero density). *Let*

$$B(s) = \sum_{j=1}^r e^{-s\tau_j} B_j, \quad B_j \in M_d(\mathbb{C}), \quad \tau_j > 0,$$

where r, d are fixed and the matrices do not depend on s . More generally, allow every entry of $B(s)$ to be a finite sum with finitely many positive real exponents and coefficients independent of s . Then

$$D(s) = \det(I - B(s))$$

is a nonzero entire exponential polynomial. For fixed real $a < b$, its zero count with multiplicity satisfies

$$\#\{D(s) = 0 : a \leq \Re s \leq b, |\Im s| \leq T\} = O(T).$$

Proof. The determinant expansion contains only finitely many products of matrix entries. Thus, for a finite set $\Lambda \subset [0, \infty)$,

$$D(s) = \sum_{\lambda \in \Lambda} c_\lambda e^{-\lambda s}.$$

Because every nonconstant roof is positive, $B(\sigma) \rightarrow 0$ as $\sigma \rightarrow +\infty$, so $D(\sigma) \rightarrow 1$. In particular, D is not identically zero.

Put $L = \max \Lambda$ and choose real σ_0 with $|D(\sigma_0)| \geq 1/2$. There is a constant C such that, on $|z| = 2R$,

$$|D(\sigma_0 + z)| \leq Ce^{2LR}.$$

Apply Jensen's formula at admissible outer radii tending down to $2R$; these radii avoid the discrete set of circles meeting a zero. The bound $|D(\sigma_0 + z)| \leq Ce^{L|z|}$ and monotone convergence give

$$\sum_{|s_k - \sigma_0| < 2R} \log \frac{2R}{|s_k - \sigma_0|} \leq \log C + 2LR - \log |D(\sigma_0)|.$$

Each zero in the inner disk of radius R contributes at least $\log 2$. Consequently

$$n_D(R) \leq \frac{\log(2C) + 2LR}{\log 2} = O(R).$$

The fixed strip rectangle of height $2T$ lies in an inner disk with $R = T + O(1)$. $\square$

Corollary 4.2 (Riemann divisor exclusion). *No fixed determinant in Theorem 4.1, no finite product or quotient of such determinants, and no product with a zero-free entire factor has the full nontrivial divisor of ξ under a fixed nonconstant complex affine spectral identification.*

Proof. A finite product or quotient, after cancelling common factors, has total zero-plus-pole count $O(T)$ in every disk of radius $O(T)$. Multiplication by a zero-free entire factor does not change the divisor. A fixed nonconstant complex affine map sends the target critical rectangle of height T into such a disk. In the standard positive-height normalization, the Riemann–von Mangoldt formula gives

$$N_\xi(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T),$$

which is not $O(T)$. The symmetric count for $|\Im s| \leq T$ is twice this quantity up to $O(1)$, so the normalization does not affect the contradiction. $\square$

When all roofs lie in $h\mathbb{N}$, the determinant is a polynomial in e^{-hs} . Each nonzero polynomial root generates an arithmetic vertical sequence with spacing $2\pi/h$. For a finite irrational roof alphabet, D is no longer periodic, but the Jensen bound remains unchanged. Thus nonlattice finite roofs remove a visible symptom, not the divisor-growth obstruction.

Remark 4.3 (Excluded objects). The theorem assumes one fixed finite-state, finite-memory model with a constant finite-dimensional twist. It does not cover a general Hölder roof represented by an infinite-rank transfer operator; infinitely many roof values; Gamma or Dirichlet structure inserted through s -dependent coefficients; nonlinear reparameterization; or a tower whose dimension and exponential type grow with the height at which zeros are counted. The next section analyzes one such tower separately.

5 Branch-pole return in a Heisenberg congruence tower

Let $q_m = 3^m$ and consider the four-regular Cayley graphs

$$X_m = \text{Cay} \left(H(\mathbb{Z}/q_m\mathbb{Z}), \{x, x^{-1}, y, y^{-1}\} \right).$$

Let B_2 be the two-loop bouquet labelled by x, y . Each $X_m \rightarrow B_2$ is a regular voltage cover with deck group $G_m = H(\mathbb{Z}/q_m\mathbb{Z})$. Separately, reduction modulo q_{m-1} gives the regular level cover $X_m \rightarrow X_{m-1}$. The Fourier/Artin decomposition over G_m contains the adjacency block

$$A_\rho = \rho(x) + \rho(x)^* + \rho(y) + \rho(y)^*$$

for each irreducible unitary ρ . A sector is *new at level m* if ρ does not factor through G_{m-1} .

The adjacency eigenvalue λ of a four-regular block contributes a Bass factor

$$1 - \lambda u + 3u^2.$$

The graph Ramanujan condition requires every nontrivial adjacency eigenvalue to lie in $[-2\sqrt{3}, 2\sqrt{3}]$. We use this graph-theoretic condition only as a pole-circle diagnostic; it is not the classical Riemann hypothesis.

5.1 Trivial-only subtraction already fails

Proposition 5.1 (A conductor-new abelian branch). *The character*

$$\chi_m(x) = e^{2\pi i/q_m}, \quad \chi_m(y) = 1$$

is new at level m , and its adjacency eigenvalue is

$$\lambda_m^{\text{ab}} = 2 + 2 \cos(2\pi/q_m) \longrightarrow 4.$$

For every $m \geq 2$, $\lambda_m^{\text{ab}} > 2\sqrt{3}$. The two roots of its Bass factor are real and converge to $1/3$ and 1 .

Proof. A one-dimensional character of frequency a factors through level $m-1$ exactly when $3 \mid a$; frequency one is new. Summing the four generator phases gives the displayed eigenvalue. At $m=2$, $2 + 2 \cos(2\pi/9) > 2\sqrt{3}$, and the sequence increases to four. The roots

$$u_{\pm, m} = \frac{\lambda_m^{\text{ab}} \pm \sqrt{(\lambda_m^{\text{ab}})^2 - 12}}{6}$$

therefore converge to the roots of $1 - 4u + 3u^2$. □

Proposition 5.1 proves that deleting only the exact trivial representation does not create a uniform spectral gap. One could respond by deleting all abelian sectors. The primitive Schrödinger blocks show that this stronger repair also fails.

5.2 A primitive nonabelian branch

The Schrödinger and Harper objects below are standard [2, 3]. Our input is the conductor-resolved Rayleigh certificate and its use as a new-sector-gap obstruction.

Lemma 5.2 (Exact-conductor Schrödinger Artin sector). *Put $\omega_m = e^{-2\pi i/q_m}$, and on $\mathbb{C}^{\mathbb{Z}/q_m\mathbb{Z}}$ define*

$$(U_m f)(t) = f(t+1), \quad (V_m f)(t) = \omega_m^t f(t).$$

Then $U_m V_m = \omega_m V_m U_m$, and

$$\rho_m(a, b, c) = \omega_m^c V_m^b U_m^a$$

is an irreducible unitary representation of G_m . It is new at level m , and its B_2 -cover adjacency block is

$$H_m = U_m + U_m^* + V_m + V_m^*.$$

This block remains in the level-new Artin quotient after all factors that descend from G_{m-1} cancel.

Proof. Using $U_m^a V_m^{b'} = \omega_m^{ab'} V_m^{b'} U_m^a$,

$$\rho_m(a, b, c) \rho_m(a', b', c') = \omega_m^{c+c'+ab'} V_m^{b+b'} U_m^{a+a'},$$

which is exactly the group law $(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab')$. Thus ρ_m is a unitary representation with

$$\rho_m(x) = U_m, \quad \rho_m(y) = V_m, \quad \rho_m(z) = \omega_m I.$$

An operator commuting with V_m is diagonal in its simple eigenbasis, and commuting also with the cyclic shift U_m forces all diagonal entries to be equal. The commutant is therefore scalar, so the representation is irreducible. Moreover,

$$\rho_m(z^{q_m/3}) = e^{-2\pi i/3} I \neq I,$$

whereas $z^{q_m/3}$ lies in the kernel of $G_m \rightarrow G_{m-1}$; hence ρ_m does not descend.

The displayed generator images give $A_{\rho_m} = H_m$. In the Artin factorization for $X_m \rightarrow B_2$, representations factoring through G_{m-1} reproduce exactly the representation-dependent factors of $X_{m-1} \rightarrow B_2$ and cancel in the level quotient. Since ρ_m does not factor, its determinant block remains. $\square$

Theorem 5.3 (Primitive Schrödinger branch return). *The representation carrying H_m is new at level m , and*

$$\lambda_{\max}(H_m) \rightarrow 4.$$

Hence, for all sufficiently large m , an exact-conductor nonabelian sector violates the four-regular Ramanujan bound. The two Bass roots associated with $\lambda_{\max}(H_m)$ converge to $1/3$ and 1 .

Proof. The triangle inequality gives $\|H_m\| \leq 4$. Put $L_m = \lfloor \sqrt{q_m} \rfloor$ and $n_m = 2L_m + 1$. For large m , let f_m be the normalized indicator of the consecutive residues $-L_m, \dots, L_m$. Direct calculation gives

$$\langle f_m, (U_m + U_m^*) f_m \rangle = 2 \left(1 - \frac{1}{n_m} \right)$$

and

$$\langle f_m, (V_m + V_m^*) f_m \rangle = \frac{2}{n_m} \sum_{j=-L_m}^{L_m} \cos \frac{2\pi j}{q_m} \geq 2 \cos \frac{2\pi L_m}{q_m}.$$

The sum of the lower bounds tends to four. The Rayleigh principle and the upper norm bound prove $\lambda_{\max}(H_m) \rightarrow 4$. At a finite level, equality in the upper bound would force simultaneously $U_m f = f$ and $V_m f = f$ for a maximizing unit vector. The Weyl relation would then give $f = e^{-2\pi i/q_m} f$, which is impossible. Hence $\lambda_{\max}(H_m) < 4$ at every finite level. The Bass-root statement follows by continuity of the quadratic roots. $\square$

The threshold can be made rational. At $q = 243$, choose $L = 15$. Since $\cos x \geq 1 - x^2/2$ and $\pi^2 < 10$, the Rayleigh quotient is larger than

$$\frac{60}{31} + 2 - \frac{1000}{6561} = \frac{769442}{203391}, \quad \frac{769442}{203391} - \frac{7}{2} = \frac{115147}{406782} > 0, \quad \frac{7}{2} > 2\sqrt{3}.$$

Thus level five already has a certified primitive nonabelian eigenvalue outside the Ramanujan interval. At every finite level the nontrivial eigenvalue is strictly below four, so its roots are not the trivial roots and cannot cancel against the topological factor. They only converge to the branching locations.

Theorem 5.3 rules out the uniform Ramanujan/new-sector-gap version of the registered regular-minus-trivial Heisenberg-tower proposal. It does not rule out a separately renormalized infinite determinant or subtraction scheme, and it does not address nonamenable property- τ towers.

6 Exact computational certificates

The computations test conventions and finite identities; none is used to infer an asymptotic theorem. All group products and determinants are exact. Floating-point numbers appear only in illustrative tables.

6.1 D4 representation-factorization control

Take the two-generator bouquet with directed edges (a, A, b, B) , inverse pairs (a, A) , (b, B) , and deck group

$$D_4 = \langle r, s : r^4 = s^2 = 1, srs = r^{-1} \rangle,$$

with voltages r, r^{-1}, s, s . For the two-dimensional representation

$$\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \rho(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the exact two-roof twisted Hashimoto determinant is

$$\begin{aligned} D_2(x, y) &= (y - 1)(y + 1) \\ &\quad \times (3x^2y - x^2 - y - 1) \times (3x^2y + x^2 - y + 1). \end{aligned}$$

Here x weights a, A and y weights b, B . At $x = y = u$,

$$D_2(u) = (u - 1)^2(u + 1)^2 \times (3u^2 - 2u + 1)(3u^2 + 2u + 1).$$

The direct 32×32 regular-cover determinant equals the product of the four one-dimensional factors and $D_2(u)^2$:

$$\begin{aligned} D_{\text{cover}}(u) &= (u - 1)^9(u + 1)^9(3u - 1)(3u + 1)(3u^2 + 1)^2 \\ &\quad \times (3u^2 - 2u + 1)^2(3u^2 + 2u + 1)^2. \end{aligned}$$

This exact equality checks orientation, inverse edges, and the required regular multiplicities.

6.2 Heisenberg chronology and aggregation

The producer verifies all statements in Proposition 3.3. An independent implementation does not import the producer. It enumerates all 343 elements of H_7 , finds that right multiplication by either z^3 or z^2 has exactly 49 cycles of length seven, and hence recovers

$$\det(I - q \operatorname{Reg}(z^3)) = \det(I - q \operatorname{Reg}(z^2)) = (1 - q^7)^{49}.$$

The resolved phase ledgers differ, as predicted.

6.3 Illustrative counts

For the degree-eight unit-roof D4 block, each nonzero polynomial root produces one vertical arithmetic sequence under $u = e^{-s}$. Table 1 shows the resulting positive-height count and the main Riemann–von Mangoldt term. The first column grows with a constant coefficient; the second eventually has a logarithmically growing coefficient. The table illustrates Corollary 4.2; it is not its proof.

The independent checker also applies adjacency directly to the frequency-one character on all 729 vertices of $H(\mathbb{Z}/9\mathbb{Z})$. The maximum eigenvector residual is 9.2×10^{-16} , consistent with binary roundoff; the eigenvalue formula itself is exact.

Table 1: Unit-roof D4 zero count versus the Riemann–von Mangoldt main term.

T	D4 block count	Riemann–von Mangoldt main term
10^2	126	29.0
10^3	1,273	648.6
10^4	12,732	10,143.0
10^5	127,322	138,068.6
10^6	1,273,238	1,747,145.5

7 Hilbert–Pólya assessment and limitations

The three obstructions answer different versions of the same proposed construction:

$$\begin{aligned}
 \text{canonical aggregation} &\implies \text{conjugacy data coarsened to order,} \\
 \text{fixed finite resolution} &\implies O(T) \text{ divisor count,} \\
 \text{natural amenable tower} &\implies \text{primitive branch-pole return.}
 \end{aligned}$$

None of these implications substitutes an averaged matrix for chronological dynamics. The first theorem evaluates ordered holonomy before aggregation; the second allows arbitrary fixed finite-memory transition blocks; the third uses exact-conductor representations at every level.

Route-A ruling. The primitive graph cycles and their representation phases are exact, but the clock is finite word or roof length rather than an intrinsic logarithmic prime clock. The divisor mismatch is a theorem for every fixed model in the registered class. The canonical finite determinant has no Gamma completion, Riemann functional equation, or Riemann–von Mangoldt law. Twisted nonbacktracking matrices are generally nonnormal rather than self-adjoint. The primitive Schrödinger blocks are self-adjoint, but their spectra return to the graph branching edge and do not define one compact-resolvent operator with the Riemann divisor. The resulting scoped score is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_FORMAL_HINT),$$

so this model is rejected as a Hilbert–Pólya candidate.

What remains open. The result does not exclude a nonamenable congruence tower with a uniform new-sector gap, provided that the tower also produces one canonical infinite determinant. It does not exclude nuclear transfer operators for non-locally-constant roofs, countably many intrinsic lengths, or joint archimedean/nonarchimedean flows. Those objects must still prove convergence, operator domains, divisor completeness, and the absence of hidden target data. A family of finite matrices indexed by a cutoff is not yet one spectral operator.

Novelty boundary. The regular-representation factorization, finite Heisenberg graph zetas, Heisenberg/Harper spectral blocks, and zero theory of exponential sums are established [2–4, 8, 9]. The claims of this paper are the explicit same-bigram chronology witness, the scoped finite-memory Hilbert–Pólya exclusion, and an explicit conductor-resolved Rayleigh certificate used to rule out the frozen uniform new-sector-gap proposal. This positioning is narrower than a new graph-zeta or Harper theorem and stronger than a finite numerical mismatch.

8 Conclusion

Deleting a trivial representation is an algebraic subtraction, not a guarantee that trivial dynamics has disappeared. In nonabelian voltage models, canonical subtraction coarsens same-order holonomies; finite representation resolution cannot supply the required global zero density; and the amenable Heisenberg tower recreates near-trivial poles in primitive nonabelian sectors. A viable next construction must change the structure, not the finite-group parameter. The most direct remaining tests are nonamenable arithmetic joint flows and infinite-rank determinants whose intrinsic roof spectrum and operator theory are defined before comparison with ξ .

A Exact ledgers

A.1 The cyclic bigram certificate

For the ordered alphabet (x, X, y, Y) , the complete common cyclic-bigram ledger of the two words in Proposition 3.3 is

	x	X	y	Y
x	1	0	1	1
X	0	2	1	0
y	2	0	0	0
Y	0	1	0	1

and the entries sum to ten. The row and column marginals both recover the common directed Parikh vector

$$(n_x, n_X, n_y, n_Y) = (3, 3, 2, 2).$$

Thus neither symbol incidence nor a first-order empirical transition matrix distinguishes the pair. The exact group product does.

A.2 Resolved central phases

With $\rho_k(z) = e^{-2\pi i k/7} I$, the exponent ledgers modulo seven are

k	1	2	3	4	5	6
$-3k$	4	1	5	2	6	3
$-2k$	5	3	1	6	4	2

for P and Q , respectively. Each row is a permutation of the nonzero residues, so multiplying symmetrically over every primitive central character collapses the distinction again. A single resolved sector distinguishes the pair but requires a canonical rule selecting k .

A.3 D4 one-dimensional factors

For $\rho(r) = \varepsilon_r$, $\rho(s) = \varepsilon_s$, the four unit-roof inverse factors are

ε_r	ε_s	$D_{\varepsilon_r, \varepsilon_s}(u)$
1	1	$-(u-1)^2(u+1)(3u-1)$
1	-1	$-(u-1)(u+1)(3u^2+1)$
-1	1	$-(u-1)(u+1)(3u^2+1)$
-1	-1	$-(u-1)(u+1)^2(3u+1)$

Their product times the square of the two-dimensional factor equals the direct regular-cover determinant coefficient by coefficient.

A.4 A general amenable warning

The explicit Heisenberg proof is a concrete instance of a broader mechanism. For a finitely generated amenable group, Følner indicators have adjacency Rayleigh quotient arbitrarily close to the degree. In sufficiently deep finite residual quotients, their finite supports embed without collision. After subtracting the constant component, the quotient still has nontrivial spectrum near the trivial edge because the group order tends to infinity. This observation explains why trivial-only subtraction is fragile in amenable towers. The primitive Schrödinger construction in Theorem 5.3 is stronger for the registered family: it identifies an exact-conductor nonabelian sector rather than only the full orthogonal complement of constants.

B Reproducibility

Software. The tested environment is Python 3.12.3 with SymPy 1.14.0, pinned in `requirements.txt`. The three executable entries are:

- producer: `code/voltage_zeta.py`;
- independent checker: `code/independent_check.py`;
- regression suite: `code/test_voltage_zeta.py`.

No prime table, Riemann-zero table, or evaluation of ξ is read.

Exact arithmetic. The D4 matrices have integer entries, so all determinant identities are computed over $\mathbb{Z}$. Heisenberg multiplication is modular integer arithmetic. The independent order-collapse check constructs the full right-regular permutations as explicit arrays and computes their cycle decompositions. It does not import the producer.

Illustrative floating-point output. Root arguments of the factored degree-eight polynomial, cosine values in the tower table, and the direct level-nine eigenvector residual use floating arithmetic. No theorem depends on those approximations. The nonabelian Ramanujan violation at modulus 243 has the rational lower certificate in Section 5.

Commands. From the project directory, run

```
python code/voltage_zeta.py --output results
python code/independent_check.py \
    --output results/independent_check.json
cd code
python -m unittest -v test_voltage_zeta.py
```

The expected test result is nine passing tests. The generated artifacts are:

- `results/exact_certificates.json`;
- `results/independent_check.json`;
- `results/unit_roof_density.csv`.

Claim-to-artifact map. The D4 factorization and all representation factors are stored under `d4_artin_factorization`. The word, bigram, holonomy, and central phase ledgers are under `heisenberg_order_collapse`. The abelian and Schrödinger tower rows are under `heisenberg_congruence_tower`. Each JSON object includes a claim-boundary string distinguishing exact certificates from numerical illustrations.

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